

Digilians

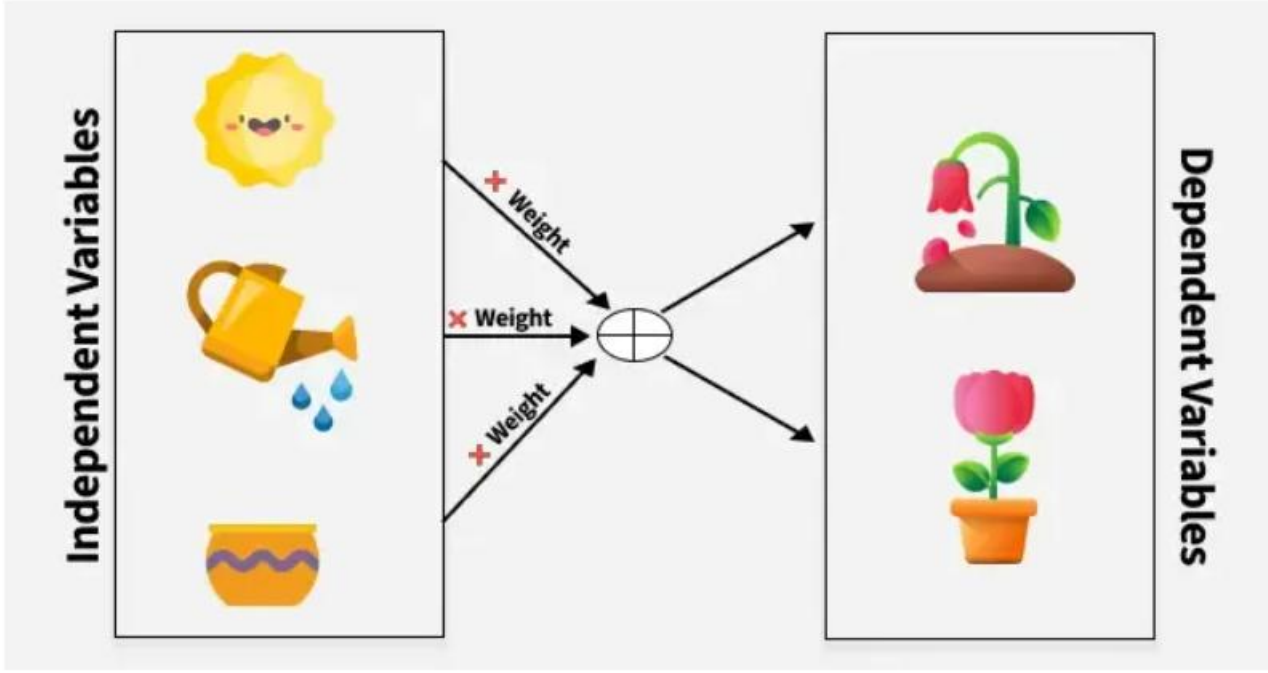
Multivariate Analysis

Multivariate Analysis

- Correlation and Covariance
- Cross-tabulation
- Cluster Analysis, MANOVA(Multivariate Analysis of Variance), Factor and Canonical Correlation Analysis

Covariance and Correlation

- Covariance and correlation are the two key concepts in Statistics that help us analyze the relationship between two variables.
- Covariance measures how two variables change together, indicating whether they move in the same or opposite directions.



- To understand this relationship better, consider factors like sunlight, water and soil nutrients (as shown in the image), which are independent variables that influence plant growth our dependent variable.
- Covariance measures how these variables change together, indicating whether they move in the same or opposite directions.

What is Covariance?

- Covariance measures how two random variables change together. It is calculated by averaging the product of their deviations from their means. A positive value means they move in the same direction, while a negative value means they move in opposite directions.
- It can take any value between - infinity to +infinity, where the negative value represents the negative relationship whereas a positive value represents the positive relationship.
- It is used for the linear relationship between variables.
- It gives the direction of relationship between variables.

Covariance Formula

1. Sample Covariance

$$\text{Cov}_S(X, Y) = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})$$

Where:

- X_i : The i^{th} value of the variable X in the sample.
- Y_i : The i^{th} value of the variable Y in the sample.
- $\bar{X}$: The sample mean of variable X (i.e., the average of all X_i values in the sample).
- $\bar{Y}$: The sample mean of variable Y (i.e., the average of all Y_i values in the sample).
- n : The number of data points in the sample.
- $\sum$: The summation symbol means we sum the products of the deviations for all the data points.
- $n - 1$: This is the degrees of freedom. When working with a sample, we divide by $n - 1$ to correct for the bias introduced by estimating the population covariance based on the sample data. This is known as Bessel's correction.

2. Population Covariance

$$\text{Cov}_P(X, Y) = \frac{1}{n} \sum_{i=1}^n (X_i - \mu_X)(Y_i - \mu_Y)$$

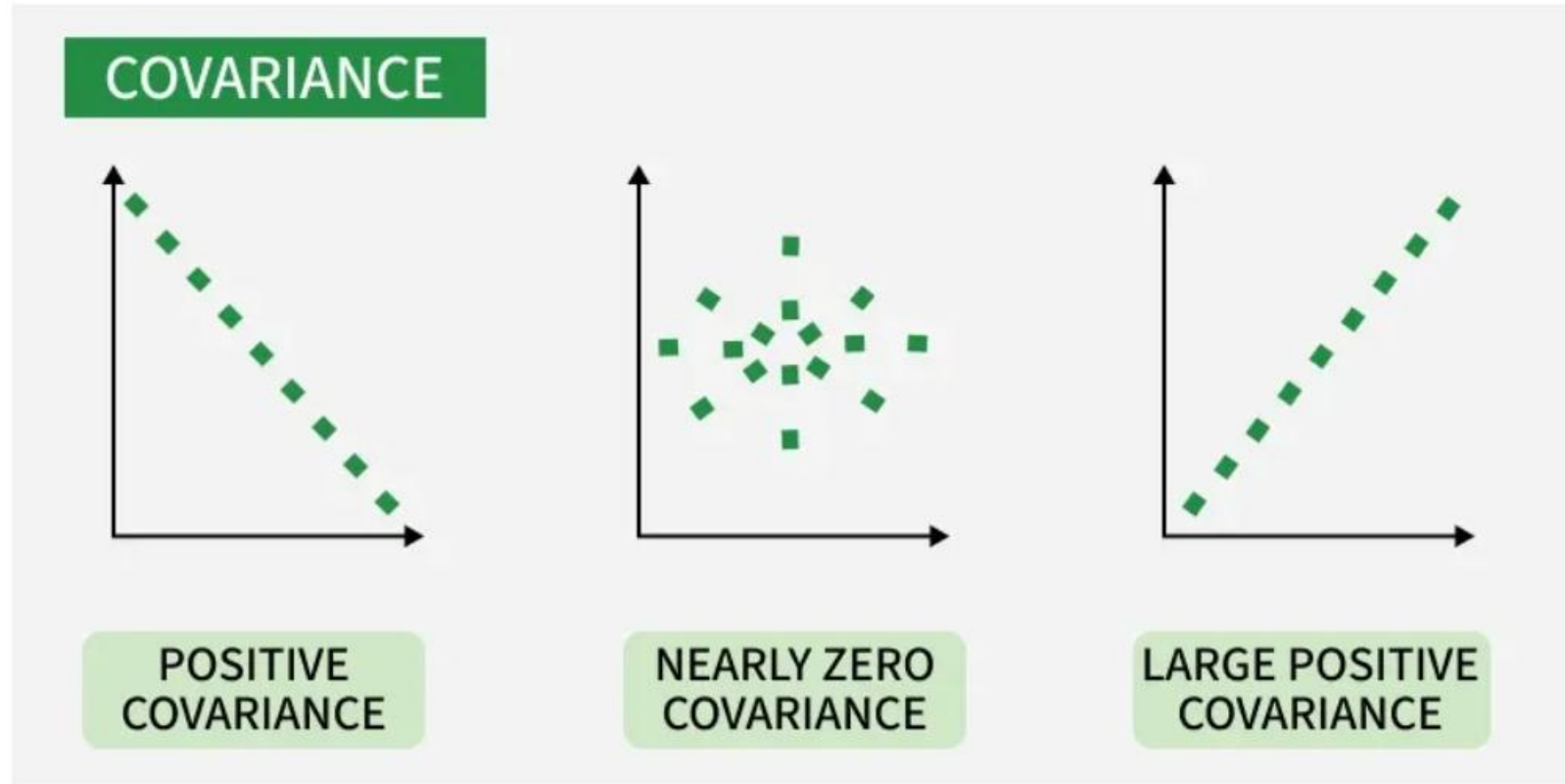
Where:

- X_i : The i^{th} value of the variable X in the population.
- Y_i : The i^{th} value of the variable Y in the population.
- μ_X : The population mean of variable X (i.e., the average of all X_i values in the population).
- μ_Y : The population mean of variable Y (i.e., the average of all Y_i values in the population).
- n : The total number of data points in the population.
- $\sum$: The summation symbol means we sum the products of the deviations for all the data points.
- n : In the case of population covariance, we divide by n because we are using the entire population data. There's no need for Bessel's correction since we're not estimating anything.

Types of Covariance

- **Positive Covariance:** When one variable increases, the other variable tends to increase as well and vice versa.
- **Negative Covariance:** When one variable increases, the other variable tends to decrease.
- **Zero Covariance:** There is no linear relationship between the two variables; they move independently of each other.

Example



Covariance

What is Correlation?

- Correlation is a standardized measure of the strength and direction of the linear relationship between two variables.
- It is derived from covariance and ranges between -1 and 1. Unlike covariance, which only indicates the direction of the relationship, correlation provides a standardized measure.

- **Positive Correlation (close to +1):** As one variable increases, the other variable also tends to increase.
- **Negative Correlation (close to -1):** As one variable increases, the other variable tends to decrease.
- **Zero Correlation:** There is no linear relationship between the variables.

- The correlation coefficient ρ for variables X and Y is defined as:
- Correlation takes values between -1 to +1, wherein values close to +1 represents strong positive correlation and values close to -1 represents strong negative correlation.
- In this variable are indirectly related to each other.
- It gives the direction and strength of relationship between variables.

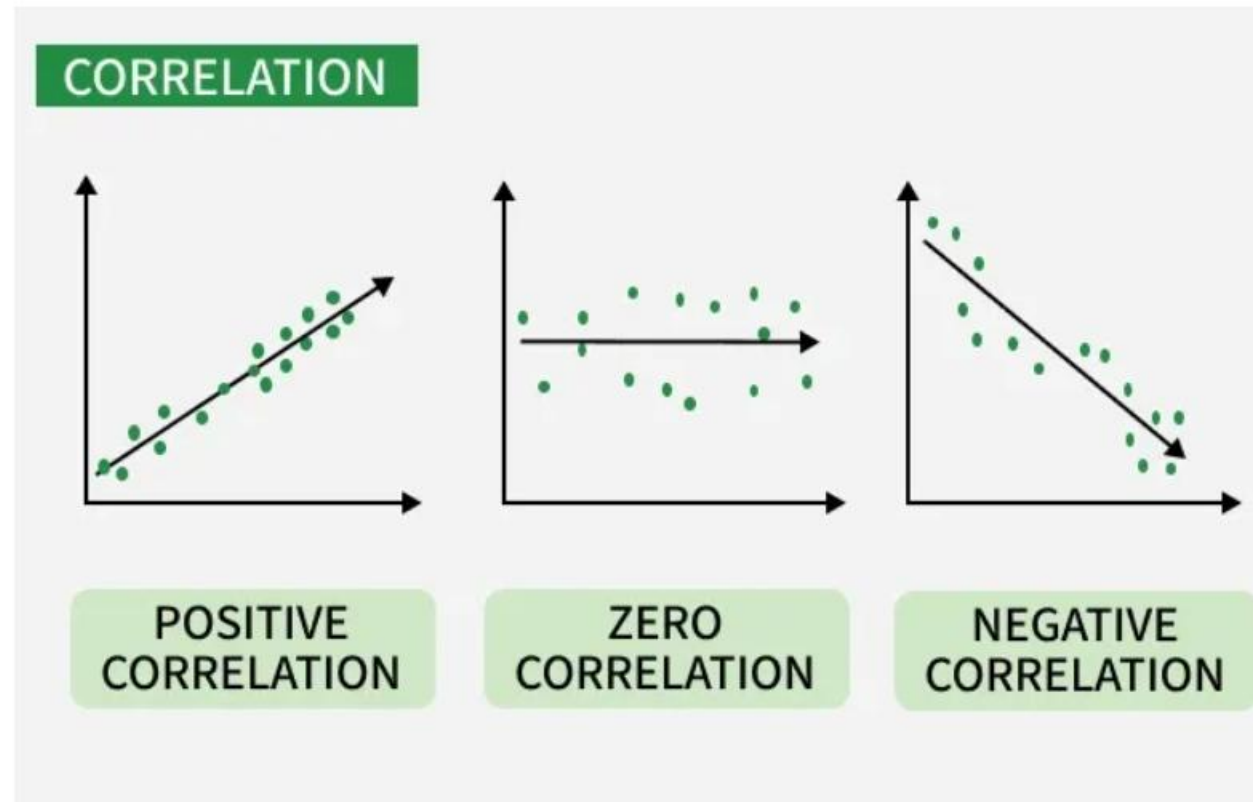
Correlation Formula

$$\text{Corr}(x, y) = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2}}$$

Here,

- $\bar{x}$ and $\bar{y}$ = mean of given sample set
- n = total no of sample
- x_i and y_i = individual sample of set

Example



Correlation

- The key difference is that Covariance shows the direction of the relationship between variables,
- while correlation shows both the direction and strength in a standardized form.

Applications of Covariance

- **Portfolio Management in Finance:** Covariance is used to measure how different stocks or financial assets move together, aiding in portfolio diversification to minimize risk.
- **Genetics:** In genetics, covariance can help understand the relationship between different genetic traits and how they vary together.
- **Econometrics:** Covariance is employed to study the relationship between different economic indicators, such as the relationship between GDP growth and inflation rates.
- **Signal Processing:** Covariance is used to analyze and filter signals in various forms, including audio and image signals.
- **Environmental Science:** Covariance is applied to study relationships between environmental variables, such as temperature and humidity changes over time.

Applications of Correlation

- **Market Research:** Correlation is used to identify relationships between consumer behavior and sales trends, helping businesses make informed marketing decisions.
- **Medical Research:** Correlation helps in understanding the relationship between different health indicators, such as the correlation between blood pressure and cholesterol levels.
- **Weather Forecasting:** Correlation is used to analyze the relationship between various meteorological variables, such as temperature and humidity, to improve weather predictions.
- **Machine Learning:** Correlation analysis is used in feature selection to identify which variables have strong relationships with the target variable, improving model accuracy.

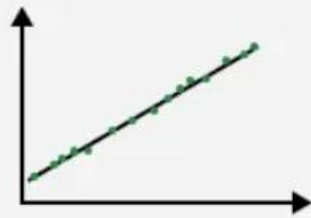
Pearson Correlation Coefficient

- **Pearson Correlation Coefficient (PCC)** is used for measuring the strength and direction of a linear relationship between two variables.
- It is important in fields like data science, finance, healthcare, and social sciences, where understanding relationships between different factors is important.
- Quantifying the degree to which two variables are related helps analysts, researchers, and decision-makers to find patterns that make predictions and inform decisions.

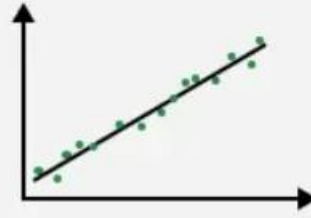
Value of PCC(denoted by r) ranges from -1 to 1:

- 1 shows a perfect positive correlation where both variables increase or decrease together at a constant rate.
- -1 shows a perfect negative correlation where one variable increases as the other decreases proportionally.
- 0 shows no linear relationship means changes in one variable do not predict changes in the other.

Pearson Correlation Coefficient



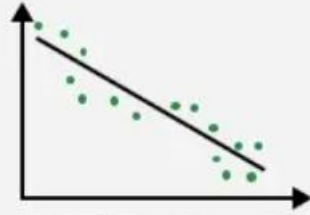
1. Strong Positive Correlation



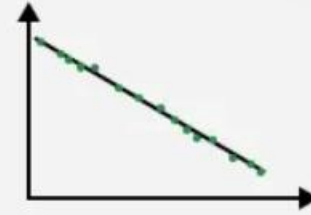
2. Medium Positive Correlation



3. Weak / No Correlation



4. Medium Negative Correlation



5. Strong Negative Correlation

Pearson Correlation Coefficient

- ***Pearson Correlation Coefficient (r)*** is known by many names such as: *Bivariate correlation, Pearson's r , Correlation coefficient, and Pearson product-moment correlation coefficient (PPMCC)*
- The PCC is a type of descriptive statistic, meaning data is presented in a meaningful way using tables, graphs, etc.

- **Assumptions of Pearson Correlation Coefficient**
- **Linear Relationship:** Pearson's correlation assumes a linear relationship between the two variables. Non-linear relationships may not be accurately captured.
- **Normality:** The variables should follow a normal distribution. This helps ensure that the correlation is meaningful and not affected by skewed data.

- **Homoscedasticity:** The variability in one variable should remain consistent across all values of the other variable. This means the spread of data should be uniform.
- **Interval or Ratio Scale:** Pearson's correlation is appropriate for interval or ratio data that are continuous and have consistent, meaningful numerical differences.
- **Independence:** Observations should be independent of each other which means one data point should not influence another. Dependent data points can distort the correlation.

Pearson's Correlation Coefficient Formula and Interpretation

The formula for Pearson's correlation coefficient is shown below:

$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n \sum x^2 - (\sum x)^2][n \sum y^2 - (\sum y)^2]}}$$

Where:

- n = number of data points
- x and y = the two variables being compared
- $\sum$ denotes summation

This formula calculates the covariance of the variables and normalizes it by their standard deviations. It provides a scale-independent measure means it is unaffected by the unit of measurement.

Pearson Correlation Coefficient Table

Pearson Correlation Coefficient (r) Range	Type of Correlation	Interpretation
$0 < r \leq 1$	Positive	Both variables increase or decrease together.
$r = 0$	None	No relationship exists between the changes in both variables.
$-1 \leq r < 0$	Negative	As one variable increases, the other decreases.

Example:

- $r = 0.85$ suggests a strong positive correlation such as more study time leading to better test scores.
- $r = -0.75$ shows a strong negative correlation like the inverse relationship between outdoor temperature and heating costs.

Pearson correlation coefficient (r) value	Strength	Direction
Greater than .5	Strong	Positive
Between .3 and .5	Moderate	Positive
Between 0 and .3	Weak	Positive
0	None	None
Between 0 and $-.3$	Weak	Negative
Between $-.3$ and $-.5$	Moderate	Negative
Less than $-.5$	Strong	Negative

Key Properties of Pearson's Correlation Coefficient

- **1. Directionality:** The sign of r shows the direction of the relationship
- Positive r means that as one variable increases, the other increases in a similar manner.
- Negative r means that as one variable increases, the other decreases.
- **2. Magnitude:** The magnitude of r shows the strength of the relationship
- Values closer to ± 1 represent a stronger relationship.
- Values closer to 0 signify a weaker or no relationship between the variables.

- **3. No Causation:** Pearson's correlation measures association but does not imply causation. Even if two variables are strongly correlated, it doesn't mean that one causes the other to change.
- **4. Symmetry:** Pearson's correlation is symmetric which mean the correlation between X and Y is the same as that between Y and X. In other words, $r(X, Y) = r(Y, X)$.
- **5. Invariance:** Pearson's correlation remains unchanged under linear transformations of the data. This means scaling (multiplying by a constant) or shifting (adding a constant) the data does not affect the correlation.

Steps to Find the Correlation Coefficient

Steps to find the correlation coefficient with Pearson's correlation coefficient formula:

Step 1: Prepare the Data: Create a table with columns for X (variable 1), Y (variable 2), XY (product of X and Y), X^2 (square of X) and Y^2 (square of Y).

Step 2: Multiply X and Y : Multiply each value of X and Y to fill the XY column.

Step 3: Square the X Values: Square each value in the X column and fill the X^2 column.

Step 4: Square the Y Values: Square each value in the Y column and fill the Y^2 column.

Step 5: Sum the Columns: Add up all the values in each column (ΣX , ΣY , ΣXY , ΣX^2 , ΣY^2).

The summation symbol (Σ) represents the sum of all the values in the respective columns.

Step 6: Apply the Formula: Now use the Pearson correlation formula:

$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n \sum x^2 - (\sum x)^2][n \sum y^2 - (\sum y)^2]}}$$

Step 7: Interpret the Result: After calculating r check whether the correlation is positive or negative based on the sign of r :

- $r > 0$ shows a positive correlation.
- $r < 0$ shows a negative correlation.
- $r = 0$ shows no linear correlation.

Pearson Correlation Coefficient Examples

Let's see some real examples for better understanding:

Example 1: There is some correlation coefficient that was given to tell whether the variables are positive or negative? **0.69, 0.42, -0.23, -0.99**

Solution:

The given correlation coefficient is as follows:

0.69, 0.42, -0.23, -0.99

Tell whether the relationship is negative or positive

- **0.69:** *The relationship between the variables is a strong positive relationship*
- **0.42:** *The relationship between the variables is a strong positive relationship*
- **-0.23:** *The relationship between the variables is a weak negative relationship*
- **-0.99:** *The relationship between the variables is a very strong negative relationship*

Example 2: Calculate the correlation coefficient for the following data by the help of Pearson's correlation coefficient formula: $X = 21, 31, 25, 40, 47, 38$ and $Y = 70, 55, 60, 78, 66, 80$

Solution:

Given variables are, $X = 21, 31, 25, 40, 47, 38$, and $Y = 70, 55, 60, 78, 66, 80$

To find the correlation coefficient of the following variables Firstly a table is to be constructed as follows, to get the values required in the formula also add all the values in the columns to get the values used in the formula.

X
21
31
25
40
47
38
$\sum x = 202$

$$\sum xy = 13937, \sum x = 202, \sum y = 409, \sum x^2 = 7280, \sum y^2 = 28265$$

Put $n = 6$ all the values in the Pearson's correlation coefficient formula:-

$$R = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n \sum x^2 - (\sum x)^2][n \sum y^2 - (\sum y)^2]}}$$

$$R = \frac{6(13937) - (202)(409)}{\sqrt{[6(7280) - (202)^2][6(28265) - (409)^2]}}$$

$$R = \frac{1004}{\sqrt{[2876][2909]}}$$

$$R = 1004 / 2892.452938$$

$$R = -0.3471$$

The correlation coefficient is -0.3471

Cross-Tabulation

- **Cross-tabulation** is also referred as crosstab. It is a statistical technique used to organize and analyze the relationship between two or more categorical variables.

What is Cross-Tabulation?

- Cross-tabulation is a special technique to organize data in table format which facilitates a clear and concise representation of the relationships between categorical variables.
- This arrangement generally involves one categorical variable to define the rows of the table and another categorical variable to define the columns where the intersections of the rows and columns contain the frequency or count of observations corresponding to the combinations of the variables.

- This tabular format allows Data Science and Machine Learning analysts and researchers to easily identify patterns, trends, and dependencies between categorical variables.

How does it organize data?

- Some of the key steps for the organizing process are listed below.
- At first, we need to load the dataset in which we are going to perform the Cross-Tabulation. This process could be easily done by Pandas module of Python.
- Now we will choose two or more categorical variables to analyze the table. These variables will be used to define the rows and columns of the cross-tabulation table.
- Now we call the 'pd.crosstab()' function to organize the table as per our pre-defined categorial variables.
- Finally, we will display the resulting cross-tabulation table and explore the relationships between the categories.



Thank You